

mass, m - a measure of the amount of matter in an object.

units kg, kilograms

distance - a measure of length. can be along one axis, x, z , etc., or have components in each, $\vec{r}$

or along a path, $\vec{s} = x\hat{x} + y\hat{y} + z\hat{z}$

units m, meters

velocity, $\vec{v}$ - change in distance with time.

$$\vec{v} = \frac{d\vec{s}}{dt} = \frac{\partial x}{\partial t} \hat{x} + \frac{\partial y}{\partial t} \hat{y} + \frac{\partial z}{\partial t} \hat{z}$$

units $\frac{m}{s}$

acceleration, $\vec{a}$ - change in velocity with time

$$\vec{a} = \frac{d\vec{v}}{dt}$$

which is also the curvature in position with time

$$\vec{a} = \frac{d^2\vec{s}}{dt^2}$$

second derivative

units $\frac{m}{s^2}$

force, $\vec{F}$ - a quantity describing the interaction between two objects, or a field and an object which causes a change in speed or direction of an object.

$$\vec{F} = m\vec{a} = m \frac{d\vec{v}}{dt} = m \frac{d^2\vec{s}}{dt^2}$$

$$N, \text{ newton} = \frac{kg \cdot m}{s^2}$$

Note: an object at rest either has no forces acting on it (unlikely) or all forces acting on it are balanced preventing motion in any direction. Gravity is pulling you towards the center of Earth. The ground exerts a force in the opposite direction balancing gravity and preventing you from falling.

Work, W - The energy expended by a force moving an object a distance in the same direction of the force.

$$W = \vec{F} \cdot \vec{s}$$

a quantity
of energy
(a scalar)

dot product indicates the projection
of the force vector in the
direction of motion vector

$$\begin{aligned} J, \text{ joule} &= N \cdot m \\ &= \frac{kg \cdot m^2}{s^2} \cdot m \\ &= \frac{kg \cdot m^3}{s^2} \end{aligned}$$

energy - there are various types of energy, which represent the potential to do work. Energy can neither be created nor destroyed, but it can be transferred and/or change form.

Types of energy used in geophysics and/or discussed in this class.

potential - energy stored due to position in a field, e.g., gravitational potential energy also magnetic and electric

kinetic - energy of motion

thermal - energy of the vibrations of atoms or molecules

nuclear - energy that binds atomic nuclei, released by radiogenic decay

chemical - energy stored in chemical bonds released or absorbed during metamorphism

elastic - energy stored in springs or in deformation of other solid (such as rocks)

ionizing - energy stored by electron energy states

magical - the energy stored in chuckles (just seeing if you're still awake)

All of these energy forms are really potential!

So how are force, work, and potential related?